The documentation and process conversion measures necessary to comply with this document shall be completed by 17 April 2024.

INCH-POUND
MIL-PRF-19500/144U
w/AMENDMENT 2
17 January 2024
SUPERSEDING
MIL-PRF-19500/144U
w/AMENDMENT 1
4 May 2023

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING, TYPES 1N4454-1, 1N4454UR-1, 1N4454UB, 1N4454UBCA, 1N4454UBCC, 1N4454UBD, 1N3064, 1N4532, JAN, JANTX, JANTXV, JANHC, AND JANKC

Device types 1N3064 and 1N4532 are inactive for new design (see 6.4).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for silicon, diffused, switching diodes. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: Axial in accordance with figure 1, surface mount version UR (DO-213AA) in accordance with figure 2, and surface mount versions UB in accordance with figure 3. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: A version die in accordance with figure 4, B version die in accordance with figure 5.

1.3 Maximum ratings. Unless otherwise specified TA = +25°C.

| Type | $V_{BR}$ | $V_{RWM}$ | $I_{O(PCB)}$ $T_A$ = 75°C (1) (2) | $I_{FSM}$ 8.3ms | $T_J$ & $T_{STG}$ | $R_{\theta JL}$ L = 3/8 inch (9.53 mm)(3) | $R_{\theta JEC}$ (UR) (3) | $R_{\theta JA(PCB)}$ (2) (3) (4) | $R_{\theta JSP}$ (UB) (3) (4) |
|---|---|---|---|---|---|---|---|---|---|
| | V dc | V (pk) | mA | A (pk) | °C | °C/W | °C/W | °C/W | °C/W |
| 1N4454-1, 1N4454UR-1 | 75 | 50 | 200 | 2 | -55 to +175 | 250 (leaded) | 100 (UR) | 325 | |
| 1N4454UB, 1N4454UBCA, 1N4454UBCC, 1N4454UBD | | | | | -55 to +200 | | | | 120 (UB) |
| 1N3064 | | | | | -55 to +175 | 250 (leaded) | | | |
| 1N4532 | | | | | | | | 325 | |

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FSC 5961

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1.3 Maximum ratings. Unless otherwise specified TA = +25°C. Continued.

(1) For temperature-current derating curves, see figure 6, and figure 7.
(2) $T_A$ = +75°C for both axial and metal electrode leadless face diodes (MELF) (UR) on printed circuit board (PCB), PCB = FR4 - .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, in still air; pads for (UR) = .061 inch (1.55 mm) x.105 inch (2.67 mm); pads for axial = .092 inch (2.34 mm) diameter, strip = .030 inch (0.76 mm) x 1 inch (25.4 mm) long, lead length L ≤ .187 inch (≤ 4.75 mm); $R_{\theta JA}$ with a defined PCB thermal resistance condition included, is measured at $I_O$ = 200 mA dc.
(3) See figure 8, figure 9, and figure 10 for thermal impedance curves.
(4) $R_{\theta JSP}$ refers to thermal resistance from junction to the solder pads of the UB package.

1.4 Primary electrical characteristics at $T_A$ = +25°C, unless otherwise specified.

| Limits (1) | $V_{F1}$ $I_F$ = 10 mA dc | $I_{R1}$ $V_R$ = 50 V dc | $C_O$ $V_R$ = 0 f = 1 MHz | $t_{rr}$ $I_F$ = $I_R$ = 10 mA dc $R_L$ = 100 Ω | $t_{fr}$ $V_{fr}$ = 5.0 V(pk) $I_F$ = 100 mA dc |
|---|---|---|---|---|---|
| Max | 0.8 V dc | 0.1 μA dc | 2 pF | 4.0 ns | 30 ns |

(1) Primary electrical characteristics for surface mount devices are equivalent to the corresponding non-surface mount devices unless otherwise specified.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for supersession information, 6.5 for PIN construction example, and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: The quality level "JAN" "JANTX" and "JANTXV".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The diodes of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are as follows: "4454", "3064", and "4532".



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1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

| | |
|---|---|
| -1 | Indicates an axial through-hole mount (DO-35) package. (see figure 1) |
| | A blank first suffix symbol indicates an axial through-hole mount DO-7, or DO-34 package (see figure 1). |
| UR-1 | Indicates a surface mount, round endcap, package. (see figure 2) |
| UB | Surface Mount types with diode configuration included (see figure 3) |

1.5.4 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and “B”.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

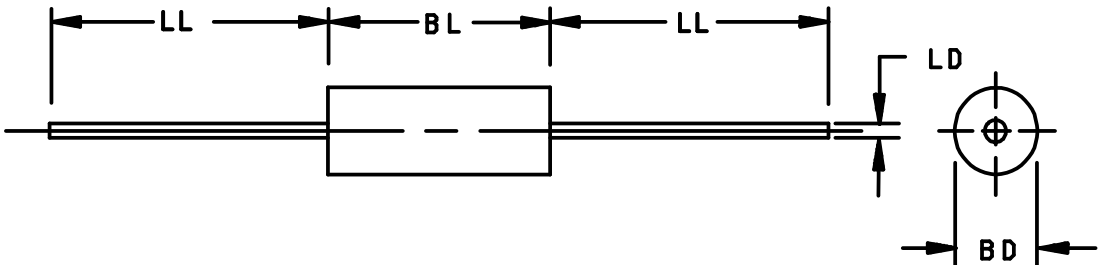
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Types | Symbol | Dimensions | | | |
|---|---|---|---|---|---|
| | | in | in | mm | mm |
| | | Min | Max | Min | Max |
| 1N4454-1 (DO-35) | BD | .056 | .075 | 1.42 | 1.91 |
| | BL | .140 | .180 | 3.56 | 4.57 |
| | LD | .018 | .022 | 0.46 | 0.56 |
| | LL | 1.000 | 1.500 | 25.40 | 38.10 |
| 1N3064 (DO-7) | BD | .078 | .107 | 1.98 | 2.72 |
| | BL | .195 | .300 | 4.96 | 7.62 |
| | LD | .018 | .022 | 0.46 | 0.56 |
| | LL | 1.000 | 1.500 | 25.40 | 38.10 |
| 1N4532 (DO-34) | BD | .050 | .075 | 1.27 | 1.91 |
| | BL | .080 | .120 | 2.03 | 3.05 |
| | LD | .018 | .022 | 0.46 | 0.56 |
| | LL | 1.000 | 1.500 | 25.40 | 38.10 |

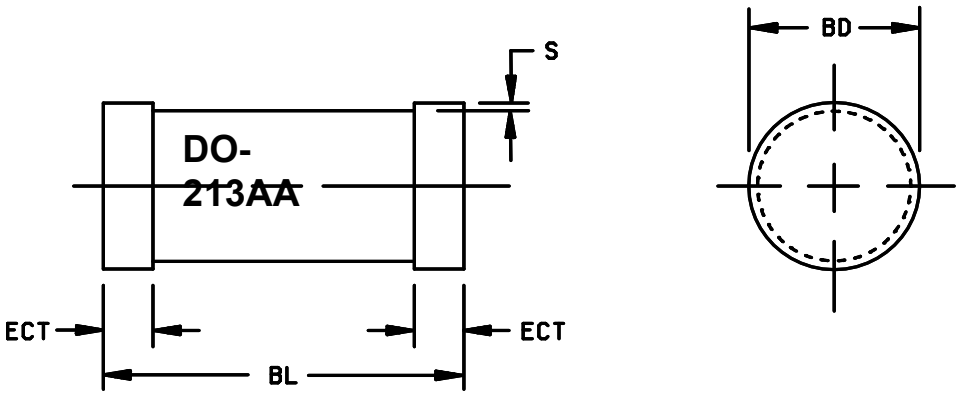
NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.

* Types 1N4454-1, 1N3064, 1N4532.

FIGURE 1. Physical dimensions.



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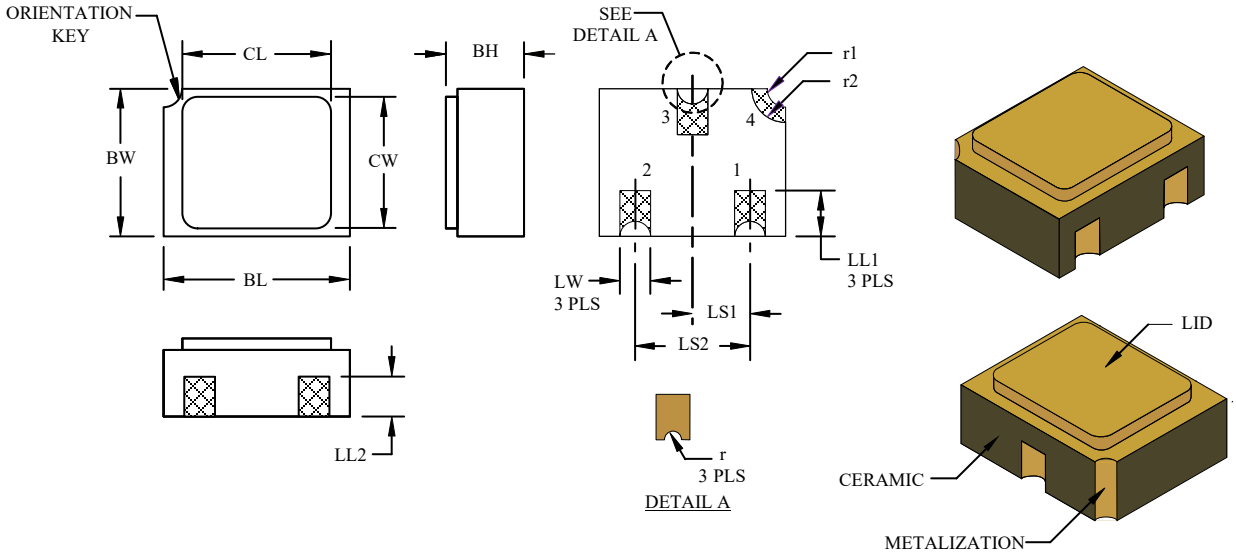

| Symbol | Dimensions | | | | |
|---|---|---|---|---|---|
| | in | In | mm | mm | Note |
| | Min | Max | Min | Max | |
| BD | .063 | .067 | 1.60 | 1.70 | |
| BL | .130 | .146 | 3.30 | 3.71 | |
| ECT | .016 | .022 | 0.41 | 0.56 | |
| S | | | | | 3 |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension S is optional however the glass body diameter shall not exceed endcap diameter.

FIGURE 2. Physical dimensions for type 1N4454UR-1 (DO-213AA).



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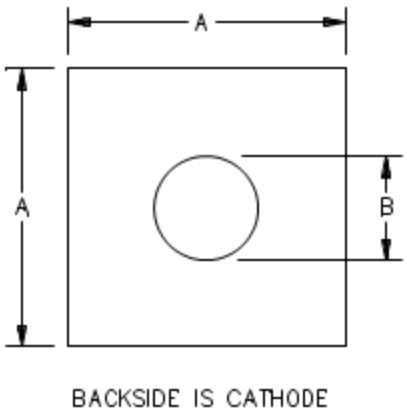
| Symbol | Dimensions | | | | Symbol | Dimensions | | | |
|---|---|---|---|---|---|---|---|---|---|
| | in | in | mm | mm | | in | in | mm | mm |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| BW | .085 | .108 | 2.16 | 2.74 | LW | .016 | .024 | 0.41 | 0.61 |
| BH | .040 | .056 | 1.02 | 1.42 | LS1 | .035 | .040 | 0.89 | 1.02 |
| BL | .115 | .128 | 2.92 | 3.25 | LS2 | .071 | .079 | 1.80 | 2.01 |
| CL | | .128 | | 3.25 | r | | .008 | | 0.20 |
| CW | | .108 | | 2.74 | r1 | | .012 | | 0.31 |
| LL1 | .022 | .038 | 0.56 | 0.97 | r2 | | .022 | | 0.56 |
| LL2 | .017 | .035 | 0.43 | 0.89 | | | | | |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.

FIGURE 3. Physical dimensions, surface mount (UB version).



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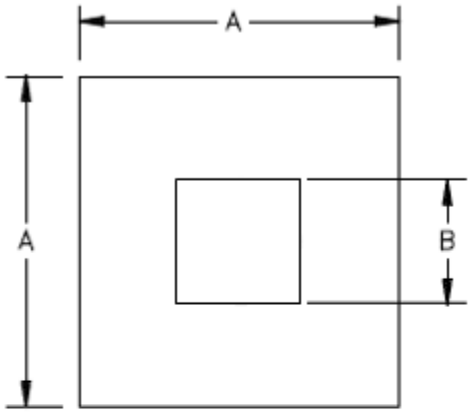

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | in | in | mm | mm |
| | Min | Max | Min | Max |
| A | .014 | .018 | .360 | .460 |
| B | .005 | .007 | .120 | .180 |
| C | .008 | .012 | 0.20 | 0.30 |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The physical characteristics of the die are:
   Metallization:
   Top (anode): Al
   Back (cathode): Au
   Al thickness: 25,000 Å minimum.
   Gold thickness: 4,000 Å minimum.
   Chip thickness: .010 inches (0.25 mm) ±.002 inches (0.05 mm).

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.



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BACKSIDE IS CATHODE




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | in | in | mm | mm |
| | Min | Max | Min | Max |
| A | .015 | .019 | .381 | .483 |
| B | .004 | .006 | .102 | .152 |
| C | .006 | .010 | .152 | .254 |

NOTES

1. Top Metal: Al-Si, 14,400 Å Minimum
2. Back Metal: Au, 5,800 Å Minimum
3. Backside: Cathode

FIGURE 5. JANHCB and JANKCB (B-version) die dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\theta JBB}$ Thermal resistance junction to burn-in board.
SP Solder pad on UB devices.
UB Hermetic unleaded 3 terminal leadless chip carrier (LCC) package type.
UR Unleaded round package type designation.
$V_{fr}$ Forward recovery voltage. Specified maximum forward voltage used to determine forward recovery time.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (axial leads), figure 2 (DO-213AA), figure 3, (UB), figure 4, and figure 5.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. All devices (except UB version) shall be metallurgically bonded, double plug construction in accordance with the requirements of MIL-PRF-19500. All glass diodes shall be designed with sufficient thermal compensation in the axial direction to optimize tensile and compressive stresses. Dimensional analysis is required of all materials used to achieve axial thermal compensation. Dimensional tolerances and corresponding coefficient of thermal expansion (CTE) shall be documented on the DSCC Design and Construction Form 36D and shall be approved by the qualifying activity to maintain qualification. Dimensional tolerances shall be sufficiently tight enough to prevent excessive stresses due to the inherent CTE mismatch. The UB devices shall be eutectically mounted and wire bonded in a ceramic package. The 'UR' version shall be structurally identical to the axial leaded versions except for end-cap lead attachment.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. The polarity shall be indicated with a contrasting color band to denote the cathode end. The prefixes JAN, JANTX, and JANTXV may be abbreviated as J, JX, and JV, respectively. The part number may be reduced to J4454, JX4454, or JV4454. No color coding shall be permitted for part numbering.

3.5.1 UR devices. For 'UR' version devices only, all marking, except polarity, may be omitted from the body, but shall be retained on the initial container. Polarity marking of 'UR' devices shall consist as a minimum, a band or three contrasting dots around the periphery of the cathode.

3.5.2 UB devices. 'UB' devices do not require polarity marking.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices Diode, Silicon, Switching, Types shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screening | Measurements for JANTXV and JANTX levels |
|---|---|
| (1) 3c | Thermal impedance (see 4.3.3) |
| 9 | Not required |
| 10 | Method 1038 of MIL-STD-750, condition A |
| (2) 11 | $I_{R1}$ and $V_{F1}$ |
| 12 | See 4.3.2 |
| (3) (4) 13 | Subgroup 2 of table I herein; $\Delta I_{R1}$ = 100 percent of initial value or 25 nA dc, whichever is greater; $\Delta V_{F1}$ = 25 mV dc. |
| 14a<br>(5) 14b | Not applicable<br>Required |

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Test within 24 hours after removal from test.
(3) When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.
(4) PDA ≤ 5 percent.
(5) For clear glass diodes, the hermetic seal (gross leak) test may be performed any time after temperature cycling, fine and gross leak required for UB package.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500 "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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4.3.2 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.2): Method 1038 of MIL-STD-750, condition B. $V_R$ = rated $V_{RWM}$; f = 50 - 60 Hz; $I_{O(min)} = I_F = I_{O(PCB)}$. $T_A$ = 75°C maximum. The maximum current density of small die shall be submitted to the qualifying activity for approval. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, mounting conditions, etc.) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining $I_H$ and $I_M$. $t_{MD}$ shall be 70 μs maximum, $t_H$ shall be 10 ms maximum. The thermal impedance limit shall comply with the thermal impedance graphs on figure 8, figure 9, and figure 10 (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3101 or 4081 of MIL-STD-750, as applicable. See group E, subgroup 4 of table II herein.

4.3.4 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements. Electrical testing shall be in accordance with table I, subgroup 2 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, table I herein, and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 herein.

4.4.2.1 Group B inspection, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Leaded samples from the same lot may be used in lieu of 'UR' suffix sample for life test.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1056 | 0°C to +100°C, 10 cycles. |
| B2 | 1051 | -55°C to +175°C, 45 cycles, including screening. |
| B2 | 2005 | $I_F$ = 100 mA, axial tensile stress = 8 lbs (3.6 kg), $T_A$ = +150°C; (not applicable to UR or UB package). |
| B3 | 1027 | $V_{(pk)}$ = rated $V_{RWM}$; f = 50 - 60 Hz; $I_O$ = 200 mA dc minimum; adjust $T_A$ or $I_O$ to obtain a minimum $T_J$ of +150°C. (See 4.5.2.) |
| B4 | 2101 | Decap analysis; (Scribe and break not applicable for UB) |
| B6 | 1032 | $T_A$ = +175°C: |



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 1056 | 0°C to + 100°C, 10 cycles. |
| C2 | 1051 | -55°C to + 175°C, 45 cycles including screening. |
| C2 | 2036 | Tension - test condition A; weight = 10 pounds (4.54 kg), t = 15 s; lead fatigue = condition E (not applicable to 'UR' and 'UB' suffix types). |
| C5 | 4081 | L = .375 inch (9.53 mm), $R_{\theta JL}$ = 250°C/W maximum; $R_{\theta JEC}$ = 100°C/W; (see 4.3.3), 22 devices, c = 0. |
| C6 | 1026 | 1,000 hours minimum, $V_{(pk)}$ = rated $V_{RWM}$; f = 50 - 60 Hz; $I_O$ = 200 mA dc minimum; adjust $T_A$ or $I_O$ to obtain a minimum $T_J$ of +150°C. (See 4.5.2.) |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each device under test still sees the full $P_t$ (minimum) and that the minimum applied voltage, where applicable, is maintained throughout the burn-in period. Method 3100 of MIL-STD-750 shall be used to measure $T_J$.

4.5.3 Forward recovery voltage and time. Forward recovery shall be measured as the time interval between zero time and the point where the pulse has decreased to 110 percent of the steady-state value of $V_F$ when $I_F$ = 100 mA dc. The maximum rise time of the response detector shall be 1 ns.



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TABLE I. Group A inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 4/ | 3101 | See 4.3.3 | $Z_{\theta JX}$ | | | °C/W |
| Forward voltage | 4011 | Condition B, $I_F$ = 10 mA dc (pulsed, see 4.5.1) | $V_{F1}$ | | .8 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 5 μA dc | $V_{BR1}$ | 50 | | V dc |
| Reverse current | 4016 | Condition A or B, DC method, $V_R$ = 50 V dc | $I_{R1}$ | | 100 | nAdc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | Condition A or B, DC method, $V_R$ = 50 V dc | $I_{R2}$ | | 100 | μA dc |
| Forward voltage | 4011 | Condition B, $I_F$ = 10 mA dc (pulsed, see 4.5.1) | $V_{F2}$ | | .7 | V dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Breakdown voltage | 4021 | $T_A$ = -55°C $I_R$ = 10 μA dc | $V_{BR2}$ | 75 | | V dc |
| Subgroup 4 | | | | | | |
| Capacitance | 4001 | $V_R$ = 0 V dc, f = 1 MHz, $V_{sig}$ = 50 $mV_{p-p}$ maximum | C | | 2.0 | pF |
| Reverse recovery time | 4031 | Condition A, $I_F$ = $I_{RM}$ = 10 mA dc | $t_{rr}$ | | 4 | ns |
| Scope display evaluation | 4023 | See method 4023 of MIL-STD-750, figures 4023-3, -7, -9, -10 only | | | | |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 6 | | | | | | |
| Surge current | 4066 | Condition A (sine wave), $i_{f(surge)}$ = 2 A(pk), $I_O$ = 0, $V_{RM}$ = 0, 10 surges, 8.3 ms width each, one surge per minute, $T_A$ = +25°C<br>or<br>Condition B (square wave), $I_{F(surge)}$ = 4 A (pk) 10 surges, 1μs width each, duty factor = 0.0055 percent, $T_A$ = 25°C | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 7 | | | | | | |
| Forward recovery voltage and time | 4026 | $I_F$ = 100 mA dc, $t_r \leq$ 1 ns (see 4.5.3.) | $V_{peak}$<br>$t_{fr}$ | | 5.0<br>30 | V (pk)<br>ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ UBCA, UBCC, and UBD devices are to have each diode tested individually.
3/ Electrical characteristics for all surface mount versions are identical to the corresponding axial leaded versions unless otherwise specified.
* 4/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.



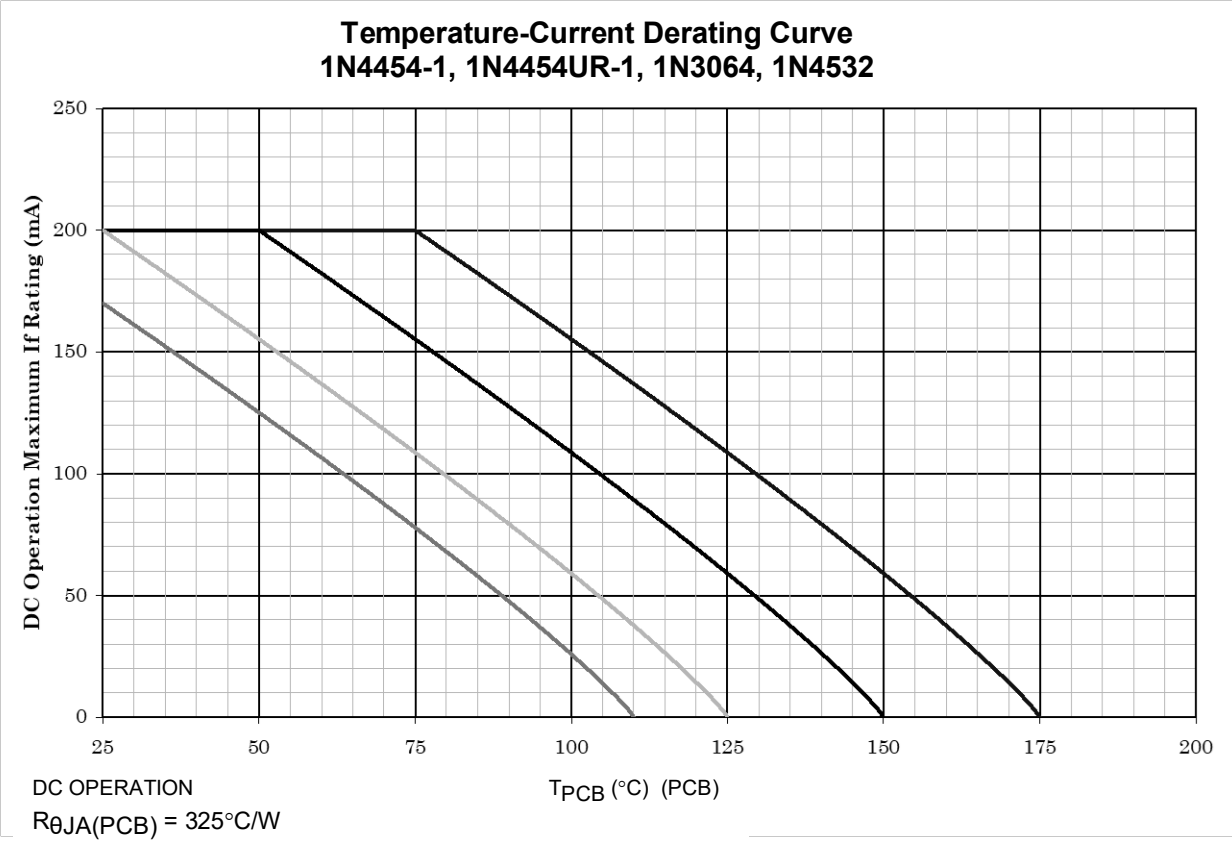
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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Qualification inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | n = 45, c = 0 |
| Thermal shock (glass strain) | 1056 | 100 cycles 0°C to 100°C. | |
| Temperature cycling | 1051 | 500 cycles, -65°C to +175°C. | |
| Hermetic seal | 1071 | Gross leak only. Fine and gross leak required for UB. | |
| Electrical measurement | | See table I, subgroup 2. | |
| Subgroup 2 | | | n = 45, c = 0 |
| Intermittent operating life | 1037 | 10,000 cycles; $I_f$ = 300 mA dc, $t_{on}$ = $t_{off}$ = 1 minute. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 4 | | | |
| Thermal resistance | 3131 | $R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). | n = 15, c = 0 |
| Thermal impedance curves | | See MIL-PRF-19500. | Sample size N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| ESD | 1020 | | |
| Subgroup 8 | | | |
| Resistance to glass cracking | 1057 | Test condition B. Test until failure occurs or to a maximum of 25 cycles, whichever comes first. | n = 45 |
| Subgroup 9 | | | n = 22, c = 0 |
| Monitored mission temperature cycling | 1055 | Not required for UB suffix devices. | |
| Electrical measurements | | See table I, subgroup 2. | |



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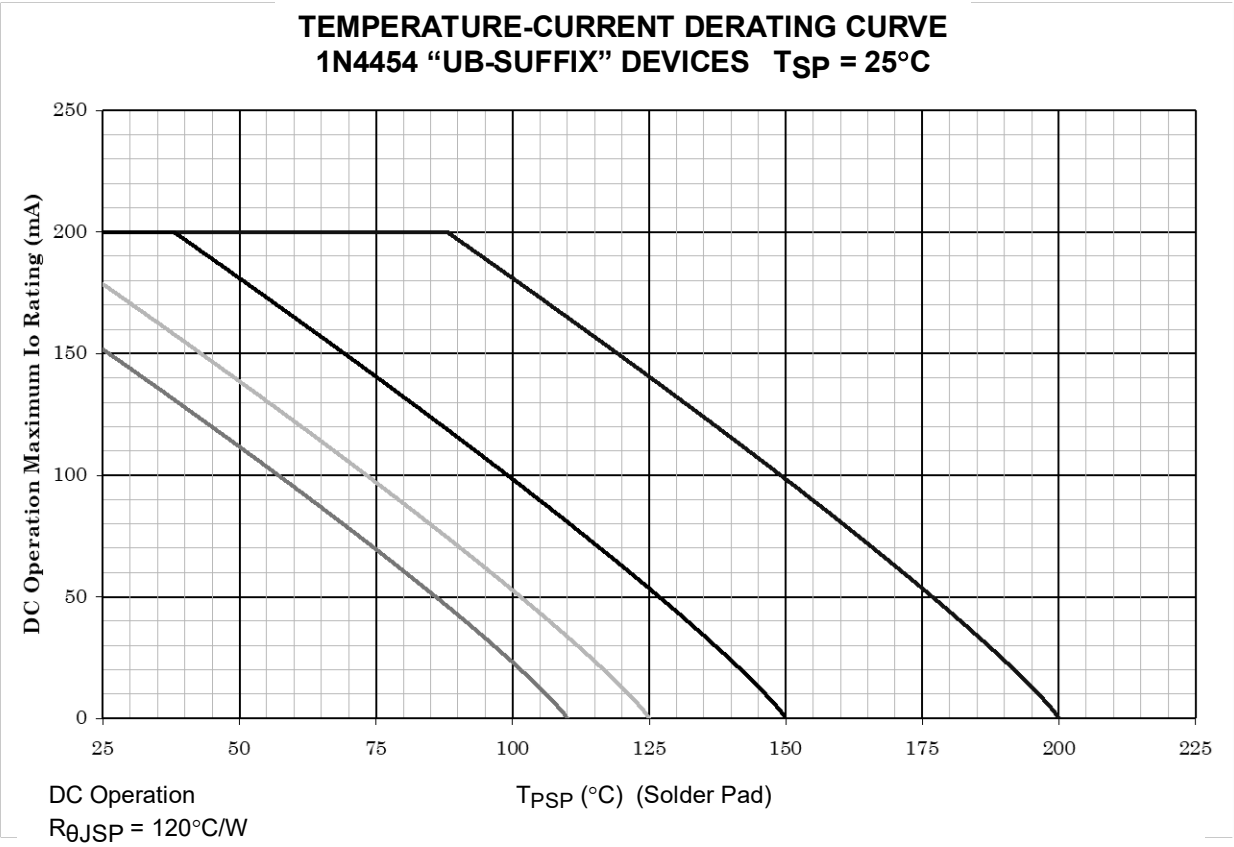

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 175°C) and current rating specified. (See 1.3.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature-current derating graph (axial and MELF).



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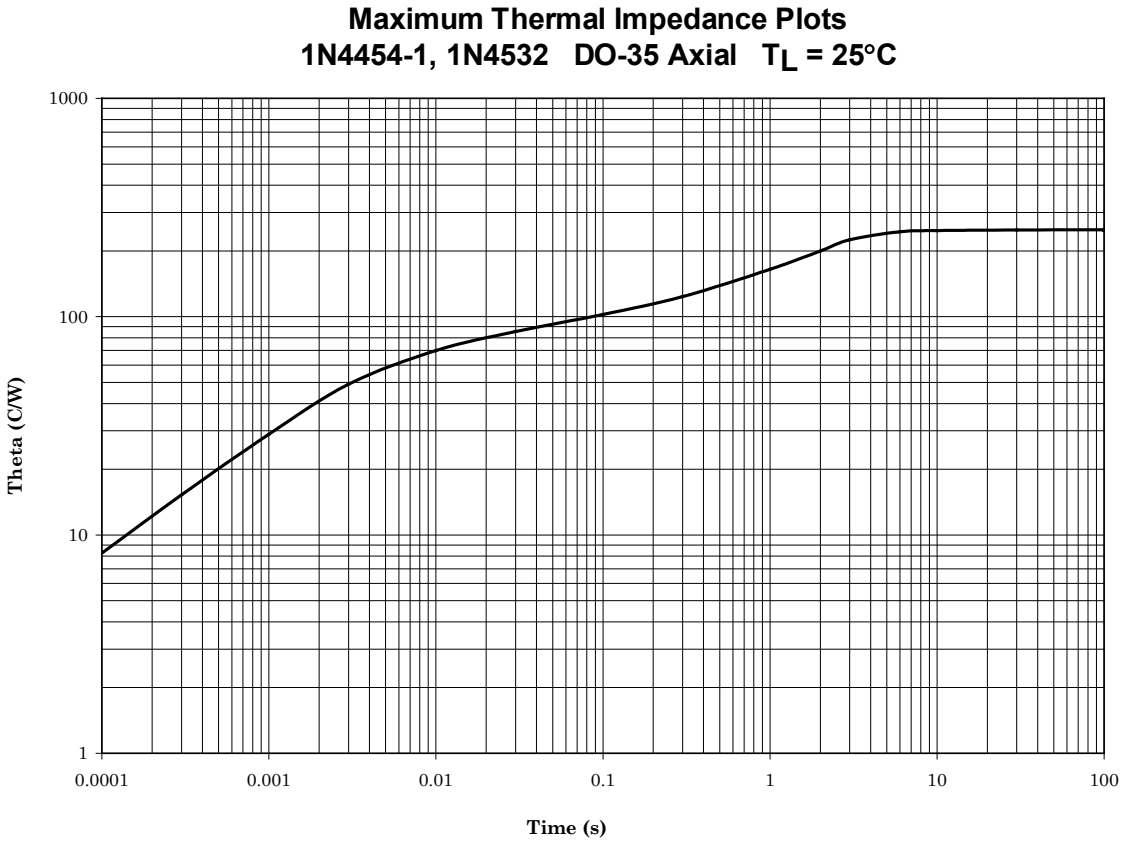

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and current rating specified. (See 1.3.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C, and +110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 7. Temperature-current derating graph ("UB-suffix" devices).



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$R_{\theta JL}$ = 250°C/W

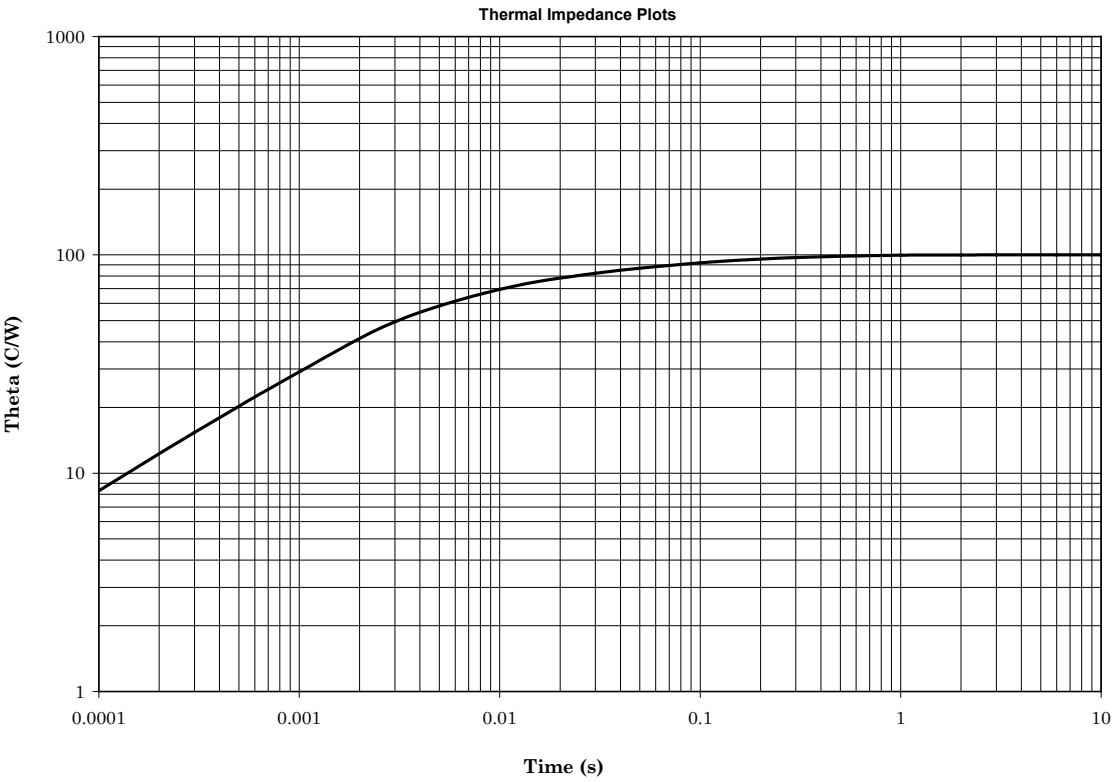
NOTE: $Z_{\theta JX}$ = 70°C/W maximum at $t_H$ = 10ms.

FIGURE 8. Thermal impedance (axial leads).



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**Maximum Thermal Impedance Plots**
**1N4454UR-1 DO-213AA $T_{EC}$ = 25°C**




$R_{\theta JEC}$ = 100°C/W

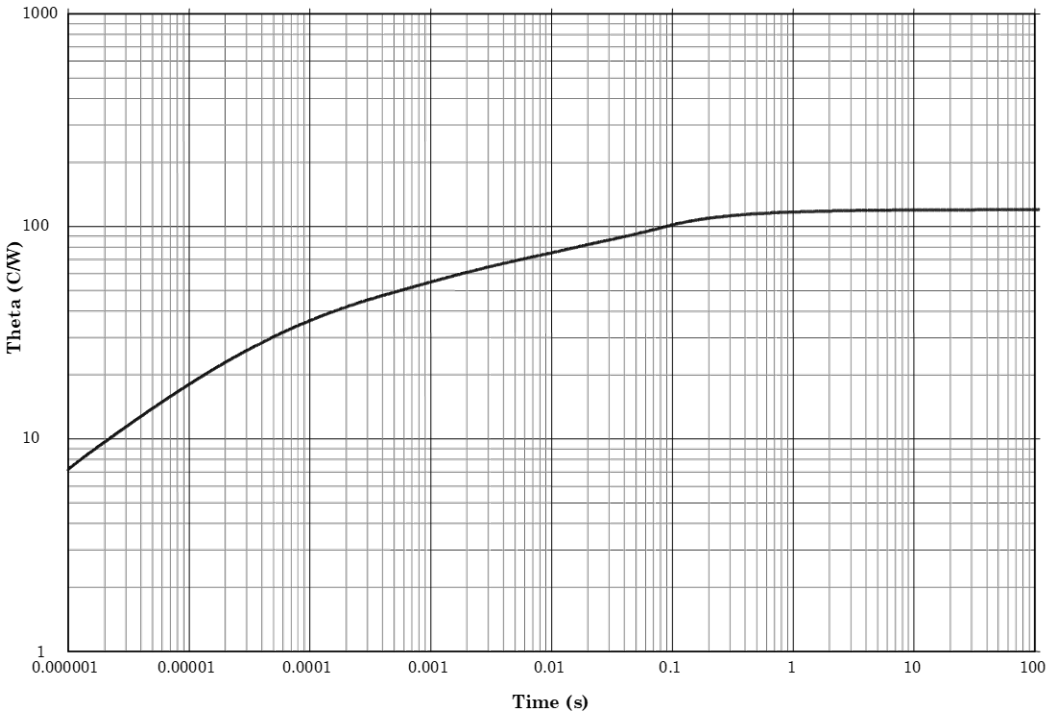
NOTE: $Z_{\theta JX}$ = 70°C/W maximum at $t_H$ = 10ms.

FIGURE 9. Thermal impedance (MELF surface mount).



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**Maximum Thermal Impedance Plots**
**1N4454UB, $T_{SP}$ = 25°C**




$R_{\theta JSP}$ = 120°C/W

NOTE: $Z_{\theta JX}$ = 90°C/W maximum at $t_H$ = 10ms.

FIGURE 10. Thermal impedance (UB versions).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5 and 6.5.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Cross reference substitution list. Dash-one devices are a direct substitute for non dash-one devices and are preferred. The 1N4454-1 is directly interchangeable for 1N3064. There will be no support for the DO-7 package. Device types 1N3064 and 1N4532 are inactive for new design. The table shows supersession information.

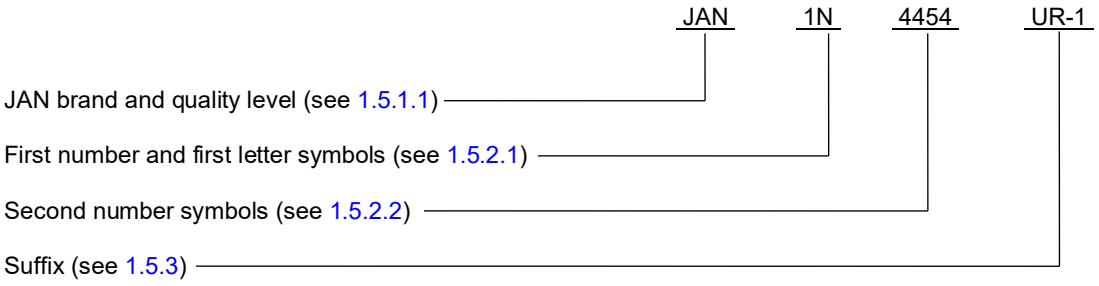
| Superseded PIN | Superseding PIN |
|---|---|
| 1N3064 | 1N4454-1 |
| 1N4454 | 1N4454-1 |
| 1N4532 | 1N4454-1 |



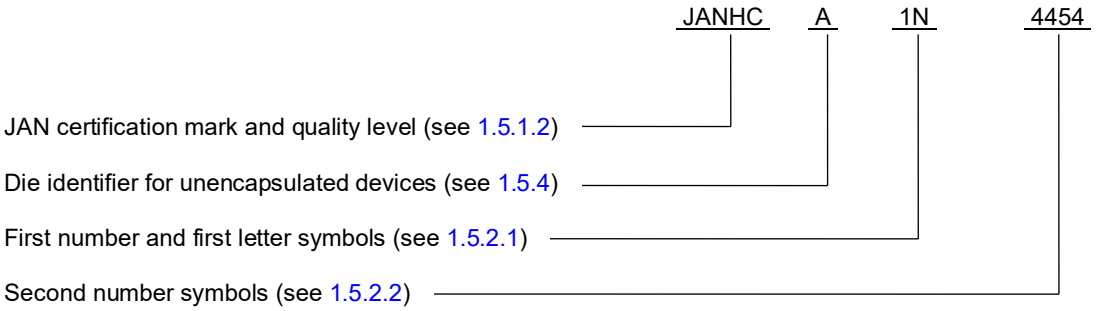
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6.5 PIN construction example.

6.5.1 PIN construction example. The PINs for encapsulated devices are constructed using the following form.




6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.




6.6 List of PINs for encapsulated devices. The following is a list of possible PINs available on this specification sheet for encapsulated devices.

| PINs for devices of the base quality level | PINs for devices of the "TX" quality level | PINs for devices of the "TXV" quality level |
|---|---|---|
| JAN1N4454-1 | JANTX1N4454-1 | JANTXV1N4454-1 |
| JAN1N4454UR-1 | JANTX1N4454UR-1 | JANTXV1N4454UR-1 |
| JAN1N4454UB | JANTX1N4454UB | JANTXV1N4454UB |
| JAN1N4454UBCA | JANTX1N4454UBCA | JANTXV1N4454UBCA |
| JAN1N4454UBCC | JANTX1N4454UBCC | JANTXV1N4454UBCC |
| JAN1N4454UBD | JANTX1N4454UBD | JANTXV1N4454UBD |
| JAN1N3064 | JANTX1N3064 | JANTXV1N3064 |
| JAN1N4532 | JANTX1N4532 | JANTXV1N4532 |



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6.6.1 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet for unencapsulated die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA1N4454) will be identified on the QML.

| JANHC and JANKC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 52GC4 |
| 1N3064 | JANHCA1N3064, JANKCA1N3064<br>JANHCB1N3064, JANKCB1N3064 |
| 1N4454 | JANHCA1N4454, JANKCA1N4454<br>JANHCB1N4454, JANKCB1N4454 |
| 1N4532 | JANHCA1N4532, JANKCA1N4532<br>JANHCB1N4532, JANKCB1N4532 |

6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2024-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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